Week 2, Model validation and evaluation: Bias-variance tradeoff

CAPP 30254, Machine Learning for Public Policy

Reminder: Our machine learning goal

Recall that we can state our machine learning goal as:

Given the training samples $(\mathbf{x}_i, y_i)$ for $i = 1, \ldots, n$ where $\mathbf{x}_i \in \mathbb{R}^p$ and $y_i \in \{+1, -1\}$, we want to learn a model that takes new data $\mathbf{x}$ and makes a correct class prediction $\widehat{y}$.

Correctly classifying new data $\mathbf{x}$ means that our model successfully *generalizes* to unseen data.

train a model: minimize $\ell(\underline{w})$

loss function

Model evaluation

Once we train a model, how do we know it works?

When we *evaluate* a model we assess the model's performance on unseen data and make sure it will work well in the real world.

Make sure it generalizes well

We can use the model's performance on the training data to estimate its overall performance. We compute the *training error* (the error from the training data) using the loss function of the model:

training error :
$$\underbrace{\frac{1}{n}\sum_{i=1}^{n}}_{\text{average}} \underbrace{\ell(y_i, \hat{y}_i)}_{\text{loss}}$$

ℓ is different for different models

Underfitting and overfitting

A low training error means that the model fits the training data well. A **very low** training error suggests *overfitting*. This happens when the model "memorizes" the training data but ~~does~~ will not generalize well to new data.

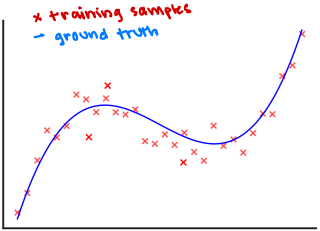
conversely,

On the other hand, a **high** training error suggests *underfitting*. An underfitted model is a model that is too simple to capture patterns in the data.

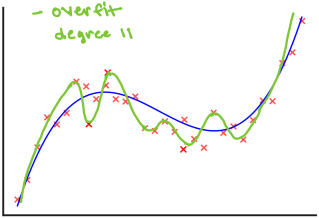
won't do well on new data

Example: Overfitting

Let's say the ground truth of the data is the curve on the left.



[1]



cubic

The overfitted model learns not only the general patterns but also the noise and specific quirks of the training set.

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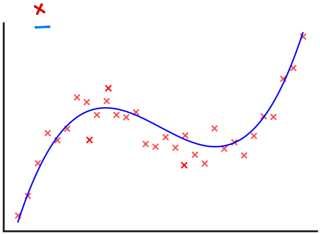
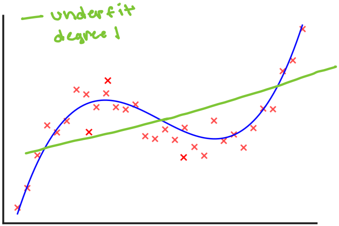
[1] Generated by ChatGPT with the prompt "could you draw a simple polynomial with some noisy data points on it?"

Example: Underfitting

Let's say the ground truth of the data is the curve on the left.



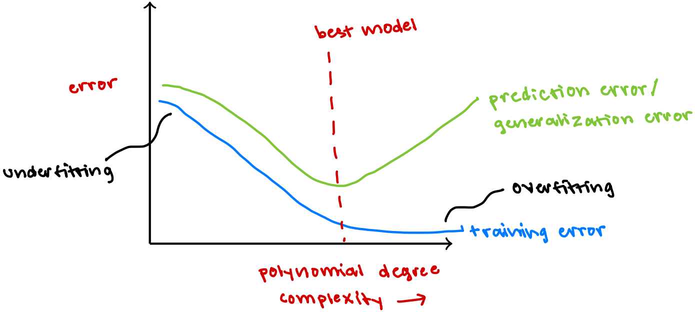


The underfitted model is too simple to capture the patterns in the training data.

doesn't find the cubic pattern

Training error

Let's graph the complexity of a model and its error.



As the training error gets very small the predictions get worse. The best model best model is somewhere before the prediction error starts to grow.

Bias and variance

Example: predict housing prices
High bias: color, # of trees, # of floors
High variance:

Machine learning models are not prefect. The *generalization error* comes from various places. Two types of error due to the complexity of a model are:

generalization error
= bias
+ variance
+ other

- Bias — High: too simple → underfitting
- Variance — too complex → overfitting

The error due to *bias* is the error introduced by approximating a real-world problem with a simplified model.

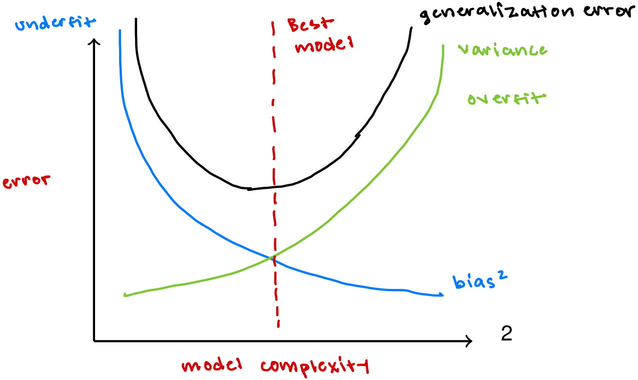
High: too simple to find actual patterns

Variance is the amount that a model's predictions would change if it where trained with a different training set. You can think of the variance as how much the model learned the noise in the training data rather than the actual underlying pattern in the data.

High: would change a lot

Bias-variance tradeoff

The *bias-variance tradeoff* is the balance between these two types of error that affect the generalization error of a model.

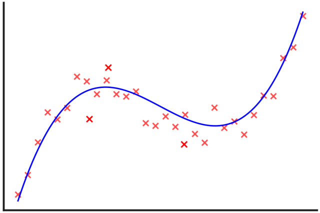


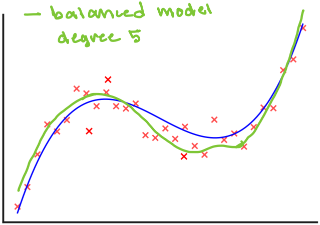
[2]

[2] https://scott.fortmann-roe.com/docs/BiasVariance.html

A balanced model

A good model should attempt minimize both bias and variance. Again, let's say the ground truth of the data is the curve on the left.



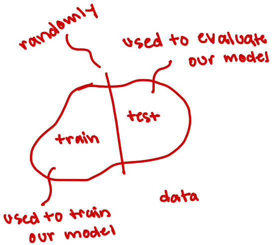


What is a better way to evaluate a model?

Using the training data to evaluate a model can be misleading. The training error tells us how well the model performs *on the data it was trained on*.

A better way to evaluate a model is to use a *test set* that is separate from our training data. We'll use the training data to train a model, then use the test set to evaluate the model after it has been trained.

Never use your test set to train your model!



[3]

[3] Generated by GPT-4, DALL-E 3 after the prompt: “Hi, could you please make a minimalistic logo for a machine learning class...” then “Could you make it less symmetrical?”